

	Requirement Tracability Matrix					
Feature	Requirement ID	Requirement Description	Test case ID	Test case description	Status	
	Functional Requirements					
Facility & Event Updates	FEU-01	The system shall display the current open/closed and availability status of facilities.	TC_FEU_01	Verify system shows current open/closed and availability status correctly	Not Run	
	FEU-02	The system shall display facilities opening and closing times for the current day.	TC_FEU_02	Verify correct opening and closing times are displayed correctly	Not Run	
	FEU-03	The system shall aggregate and display upcoming events with the event name, time and location.	TC_FEU_03	Verify system display upcoming events with name, time and location correctly	Not Run	
	FEU-04	The system shall display near real-time data.	TC_FEU_04	Verify data updates without any delay	Not Run	
	FEU-05	The system shall display last retrieved data when live data cannot fetch.	TC_FEU_05	Verify last saved data is shown when no internet	Not Run	
Transit & Navigations	TN-01	The system shall display campus bus routes.	TC_TN_01	Verify bus routes are are displayed correctly for each bus	Not Run	
	TN-02	The system shall display arrival and departure time.	TC_TN_02	Verify arrival and departure times accurately displayed for each bus	Not Run	
	TN-03	The system shall provide updated schedules.	TC_TN_03	Verify system shows latest transit shedule updates	Not Run	
	TN-04	The system shall display current status such as on time, delayed or unavailable.	TC_TN_04	Verify bus status shows 'On Time' when no delay	Not Run	
			TC_TN_05	Verify bus status shows 'Delayed' when delay exists	Not Run	
			TC_TN_06	Verify bus status shows 'Unavailable' when no busses available	Not Run	
Club Hub	CH-01	The system shall let students to discover and join active communities.	TC_CH_01	Verify student can join a club successfully	Not Run	
	CH-02	The system shall display active clubs, communities with details.	TC_CH_02	Verify all clubs are shown with correct details	Not Run	
	CH-03	The system shall feature student-led groups like coding club, cybersecurity club and etc.	TC_CH_03	Verify featured clubs section is displayed	Not Run	
	CH-04	The system shall display club activities including upcoming meetings and events.	TC_CH_04	Verify system display club activities correctly	Not Run	
Check-in & Rewards System	CRS-01	The system shall provide a check-in button or action when user is at valid campus locations.	TC_CRS_01	Verify user can check-in at valid campus locations using QR scan	Not Run	
	CRS-02	The system shall validate user location before awarding points	TC_CRS_02	Verify system validates location before awarding points	Not Run	
	CRS-03	Upon a successful validated check-in, the system shall award a defined number of engagement points (e.g. +10 points) to the user's profile.	TC_CRS_03	Verify points are received after successful check-in	Not Run	
	CRS-04	The system shall prevent a student from checking in at the same location more than once within a defined cool down period.	TC_CRS_05	Verify duplicate check-in is prevented	Not Run	

		CRS-05	Key locations for check-in include university library, cafeteria, club meetings, event venues and transport.	TC_CRS_04	Verify check-in allowed only at valid campus locations	Not Run
		CRS-06	The system shall display students current points balance.	TC_CRS_06	Verify current points balance displayed correctly	Not Run
		CRS-07	The system shall maintain and display a log of the student's past check-in events.	TC_CRS_07	Verify check-in history is available and shown	Not Run
		CRS-08	The system shall implement logic to allow points to be redeemed against defined rewards.	TC_CRS_08	Verify rewards redemption works correctly	Not Run
	User Login	LOG-01	Verify registered user can log in with valid Student ID and Password	TC_LP_01	Verify successful login with correct id and password	Not Run
		LOG-02	Verify user cannot log in with invalid credentials	TC_LP_15	Login with incorrect id and password	Not Run
		LOG-03	Verify user profile (including points balance) loads correctly after login	TC_LP_03	Verify correct points balance displayed after login	Not Run
		LOG-05	Verify session expires after inactivity	TC_LP_07	Verify session expires after inactivity	Not Run
		LOG-06	Verify login page uses secure connection (HTTPS)	TC_LP_09	Verify login uses HTTPS secure connection	Not Run
		LOG-07	Verify no sensitive data (password, tokens) appears in URL or logs	TC_LP_10	Verify no sensitive data in URL or logs after login	Not Run
		LOG-08	Verify account lock after multiple failed login attempts	TC_LP_08	Verify account lock after multiple failed attempts	Not Run
		LOG-09	Verify password is case sensitive	TC_LP_05	Verify password is case sensitive	Not Run
		LOG-10	Verify password is masked during entry	TC_LP_04	Verify password is masked	Not Run
		LOG-11	Verify SQL injection is prevented in login fields	TC_LP_17	SQL injection in Student ID field	Not Run
				TC_LP_18	SQL injection in Password field	Not Run
	User Registration	REG-01	Create a new student user profile with valid student ID and email	TC_RP_01	Register with valid information	Not Run
		REG-02	Attempt to create a duplicate profile (same student ID)	TC_RP_17	Verify duplicate student id rejected	Not Run
		REG-03	Create profile with missing required fields	TC_RP_19	Missing required fields	Not Run
		REG-04	Create profile with invalid email format	TC_RP_14	Invalid email format	Not Run
		REG-05	After registration, verify data is encrypted/stored securely	TC_RP_05	Verify password is stored encrypted	Not Run
		REG-06	After registration, verify user starts with 0 engagement points	TC_RP_04	Verify new user starts with 0 points	Not Run
	Non-Functional Requirements					
	User Profiles & Authentication	UPA-01	The system must be able to create and manage student user profiles.	TC_UPA-01	Verify profile creation after registration	Not Run
		UPA-02	The system must implement measures to ensure data integrity (e.g points balances are accurate and cannot be easily manipulated).	TC_UPA-02	Verify all data are accurate	Not Run
				TC_UPA-03	Verify data cannot be manipulated	Not Run

		UPA-03	The system shall display the student's profile including their details	TC_UPA-04	Verify student can view their profiles	Not Run
	Data Management & Security	DMS-01	All student data, including profile information and reward points balances, must be stored securely.	TC_DMS_01	Verify sensitive data stored securely	Not Run
		DMS-02	Student IDs must be masked in all display contexts. The system must not expose sensitive student data in logs or UI and protect from unauthorized users.	TC_DMS_02	Verify student id is masked in all display contexts	Not Run
				TC_DMS_03	Verify sensitive data is not exposed	Not Run
				TC_DMS_04	Verify system is protected from unauthorized users	Not Run
		DMS-03	The points balance must only be modifiable through server-side validated check-in logic, preventing client-side manipulation.	TC_DMS_05	Verify point balance cannot be manipulated	Not Run
	Reliability & Availability	RA-01	The system must handle all possible failures gracefully, without crashing or presenting unhandled exceptions to the user.	TC_RA_01	Verify system handles API failure without crash	Not Run
		RA-02	In the event of a failed data feed, the application must display the most recently cached data.	TC_RA_02	Verify recently cached data is shown	Not Run